CCNA Capstone Project

Creative Innovation Hub

Global Logistics Network

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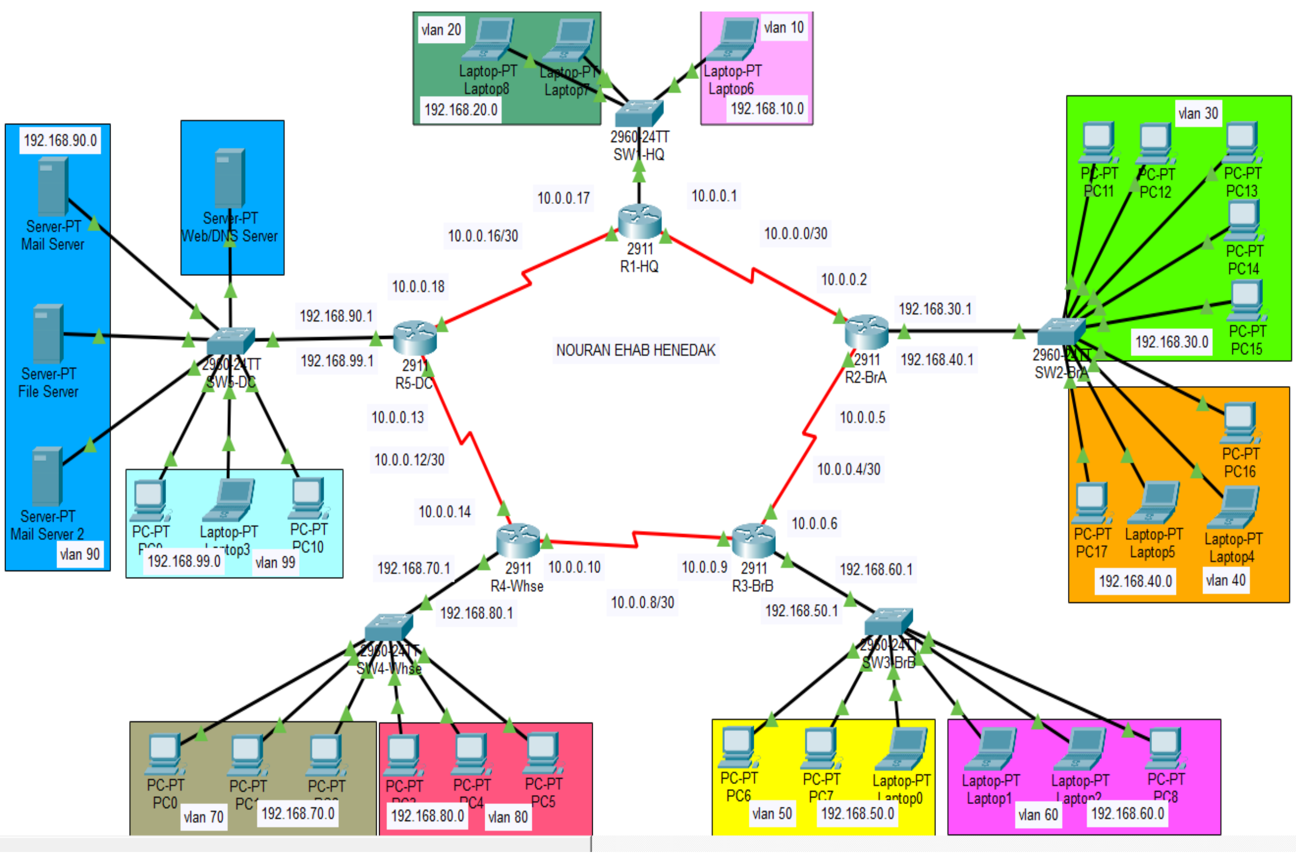
Project Overview :

The Global Logistics Network (GLN) consists of 5 interconnected sites :

- Headquarter (HQ)
- Branch A (Sales)
- Branch B (Marketing)
- Warehouse
- Data center

The network is designed using :

- VLAN segmentation
- Router-on-a-stick (inter-VLAN Routing)
- EIGRP Dynamic Routing
- DHCP services
- DNS services
- Access Control Lists (ACLs)



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IP Addressing Scheme :

site	VLAN	Name	Subnet	Gateway
HQ	10	Admin	192.168.10.0/24	192.168.10.1
HQ	20	Finance	192.168.20.0/24	192.168.20.1
Branch A	30	Sales	192.168.30.0/24	192.168.30.1
Branch A	40	Support	192.168.40.0/24	192.168.40.1
Branch B	50	Marketing	192.168.50.0/24	192.168.50.1
Branch B	60	Dev	192.168.60.0/24	192.168.60.1
Warehouse	70	Inventory	192.168.70.0/24	192.168.70.1
Warehouse	80	Shipping	192.168.80.0/24	192.168.80.1
Data Center	90	Servers	192.168.90.0/24	192.168.90.1
Data Center	99	Management	192.168.99.0/24	192.168.99.1

WAN Networks :

Link	Subnet
R1-R2	10.0.0.0/30
R2-R3	10.0.0.4/30
R3-R4	10.0.0.8/30
R4-R5	10.0.0.12/30
R5-R1	10.0.0.16/30

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Switching Configuration :

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW3-BrB
SW3-BrB(config)#enable secret class
SW3-BrB(config)#
SW3-BrB(config)#line console 0
SW3-BrB(config-line)#password cisco
SW3-BrB(config-line)#login
SW3-BrB(config-line)#
SW3-BrB(config-line)#line vty 0 4
SW3-BrB(config-line)#password cisco
SW3-BrB(config-line)#login
SW3-BrB(config-line)#
SW3-BrB(config-line)#service password-encryption
SW3-BrB(config)#
SW3-BrB(config)#vlan 50
SW3-BrB(config-vlan)#name Marketing
SW3-BrB(config-vlan)#
SW3-BrB(config-vlan)#exit
SW3-BrB(config)#
SW3-BrB(config)#interface range fa0/2 - 4
SW3-BrB(config-if-range)#switchport mode access
SW3-BrB(config-if-range)#witchport access vlan 50
                          ^
% Invalid input detected at '^' marker.

SW3-BrB(config-if-range)#switchport access vlan 50
SW3-BrB(config-if-range)#exit
SW3-BrB(config)#
SW3-BrB(config)#vlan 60
SW3-BrB(config-vlan)#name Dev
SW3-BrB(config-vlan)#
SW3-BrB(config-vlan)#interface range fa0/5 - 7
SW3-BrB(config-if-range)#switchport mode access
SW3-BrB(config-if-range)#switchport access vlan 60
SW3-BrB(config-if-range)#exit
SW3-BrB(config)#
SW3-BrB(config)#interface fa0/1
SW3-BrB(config-if)#switchport mode trunk
SW3-BrB(config-if)#exit
SW3-BrB(config)#
```

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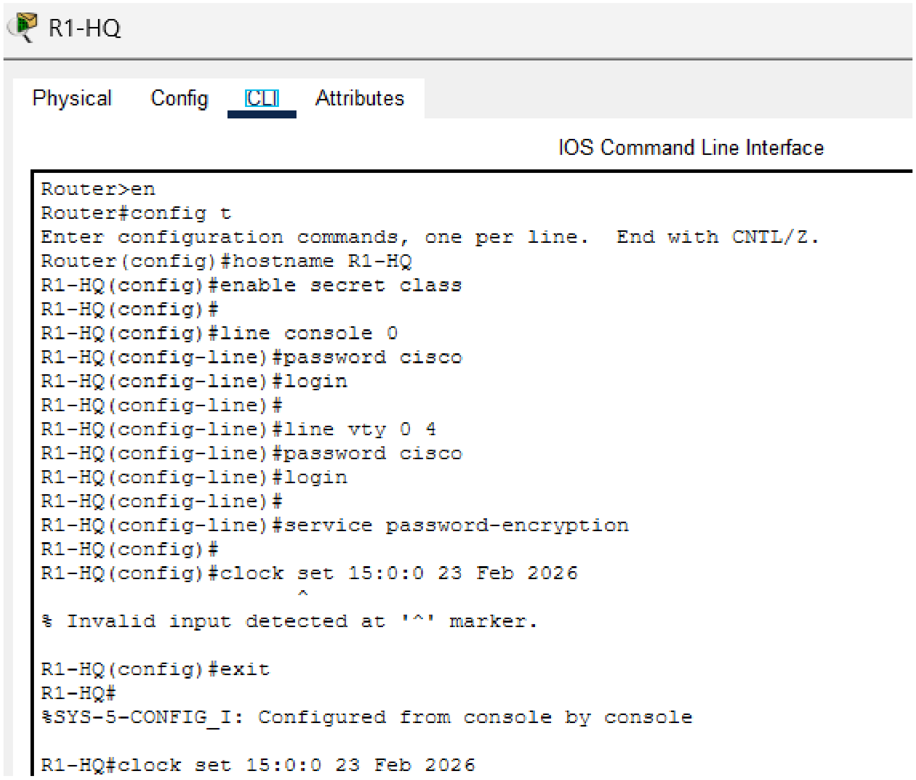
Routing Configuration :

Basic Configuration:

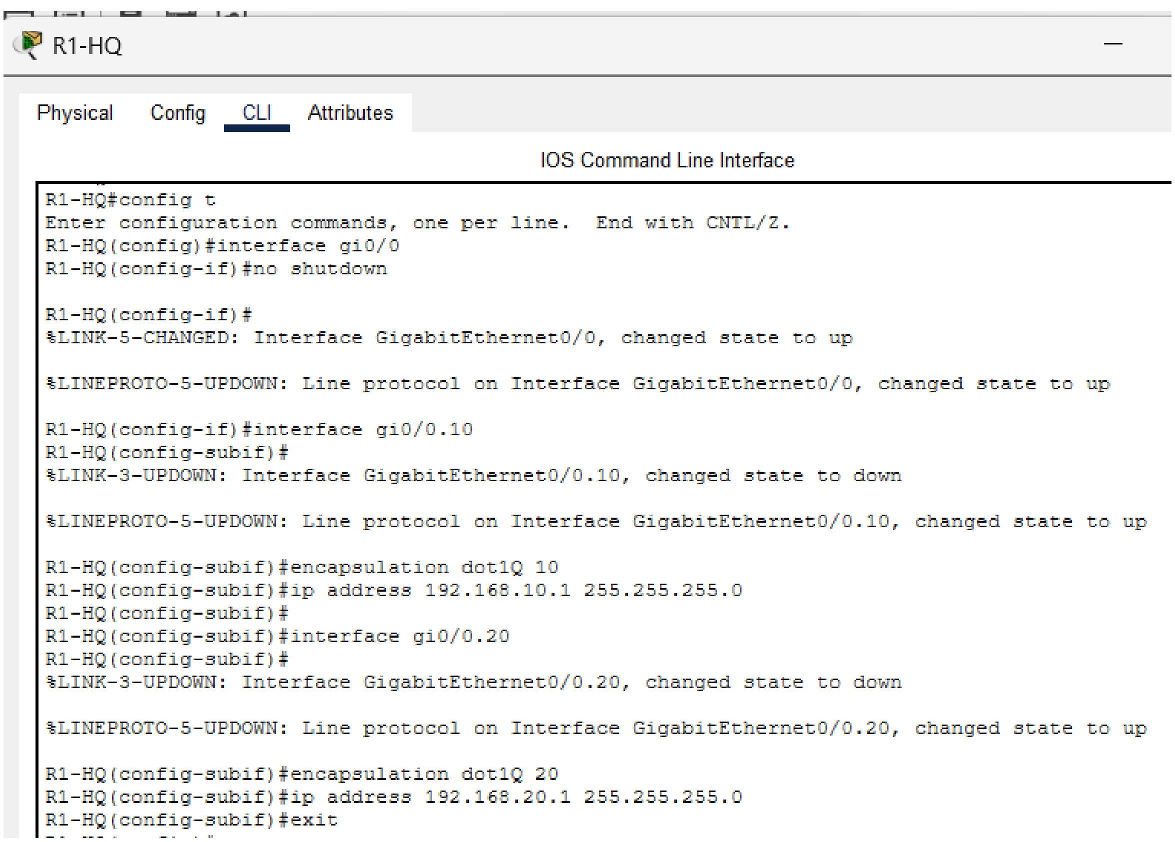



```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1-HQ
R1-HQ(config)#enable secret class
R1-HQ(config)#
R1-HQ(config)#line console 0
R1-HQ(config-line)#password cisco
R1-HQ(config-line)#login
R1-HQ(config-line)#
R1-HQ(config-line)#line vty 0 4
R1-HQ(config-line)#password cisco
R1-HQ(config-line)#login
R1-HQ(config-line)#
R1-HQ(config-line)#service password-encryption
R1-HQ(config)#
R1-HQ(config)#clock set 15:0:0 23 Feb 2026
                    ^
% Invalid input detected at '^' marker.

R1-HQ(config)#exit
R1-HQ#
%SYS-5-CONFIG_I: Configured from console by console

R1-HQ#clock set 15:0:0 23 Feb 2026
```

ROAS (Inter-VLAN Routing):




```
R1-HQ#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1-HQ(config)#interface gi0/0
R1-HQ(config-if)#no shutdown

R1-HQ(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1-HQ(config-if)#interface gi0/0.10
R1-HQ(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

R1-HQ(config-subif)#encapsulation dot1Q 10
R1-HQ(config-subif)#ip address 192.168.10.1 255.255.255.0
R1-HQ(config-subif)#
R1-HQ(config-subif)#interface gi0/0.20
R1-HQ(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

R1-HQ(config-subif)#encapsulation dot1Q 20
R1-HQ(config-subif)#ip address 192.168.20.1 255.255.255.0
R1-HQ(config-subif)#exit
```

WAN Configuration:

```
R1-HQ(config)#interface s0/0/0
R1-HQ(config-if)#ip address 10.0.0.1 255.255.255.252
R1-HQ(config-if)#clock rate 64000
R1-HQ(config-if)#no shutdown

R1-HQ(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R1-HQ(config-if)#interface s0/0/1
R1-HQ(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R1-HQ(config-if)#ip address 10.0.0.17 255.255.255.252
R1-HQ(config-if)#clock rate 64000
R1-HQ(config-if)#no shutdown

R1-HQ(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

R1-HQ(config-if)#exit
```

EIGRP Routing Configuration:

- EIGRP AS Number : 100
- Auto summary disabled

```
R1-HQ(config)#router eigrp 100
R1-HQ(config-router)#no auto-summary
R1-HQ(config-router)#network 10.0.0.0
R1-HQ(config-router)#network 10.0.0.16
R1-HQ(config-router)#network 192.168.10.0
R1-HQ(config-router)#network 192.168.20.0
R1-HQ(config-router)#exit
R1-HQ(config)#
```

DHCP Configuration:

- Each router acts as a DHCP server for its VLANs
- First 10 IP addresses excluded
- Default Gateway = .1
- DNS Server = 192.168.90.10

```
R1-HQ(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
R1-HQ(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
R1-HQ(config)#
R1-HQ(config)#ip dhcp pool POOL-Admin
R1-HQ(dhcp-config)#network 192.168.10.0 255.255.255.0
R1-HQ(dhcp-config)#default router 192.168.10.1
                           ^
% Invalid input detected at '^' marker.

R1-HQ(dhcp-config)#default-router 192.168.10.1
R1-HQ(dhcp-config)#dns-server 192.168.90.10
R1-HQ(dhcp-config)#exit
R1-HQ(config)#
R1-HQ(config)#ip dhcp pool POOL-Finance
R1-HQ(dhcp-config)#network 192.168.20.0 255.255.255.0
R1-HQ(dhcp-config)#default-router 192.168.20.1
R1-HQ(dhcp-config)#dns-server 192.168.90.10
R1-HQ(dhcp-config)#exit
R1-HQ(config)#do wr
Building configuration...
[OK]
```

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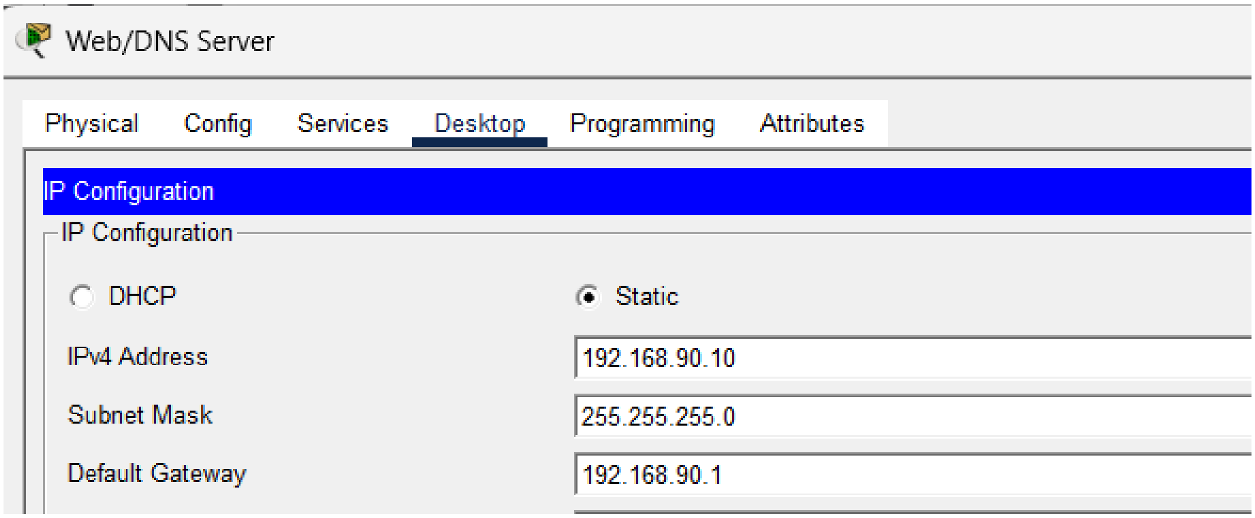
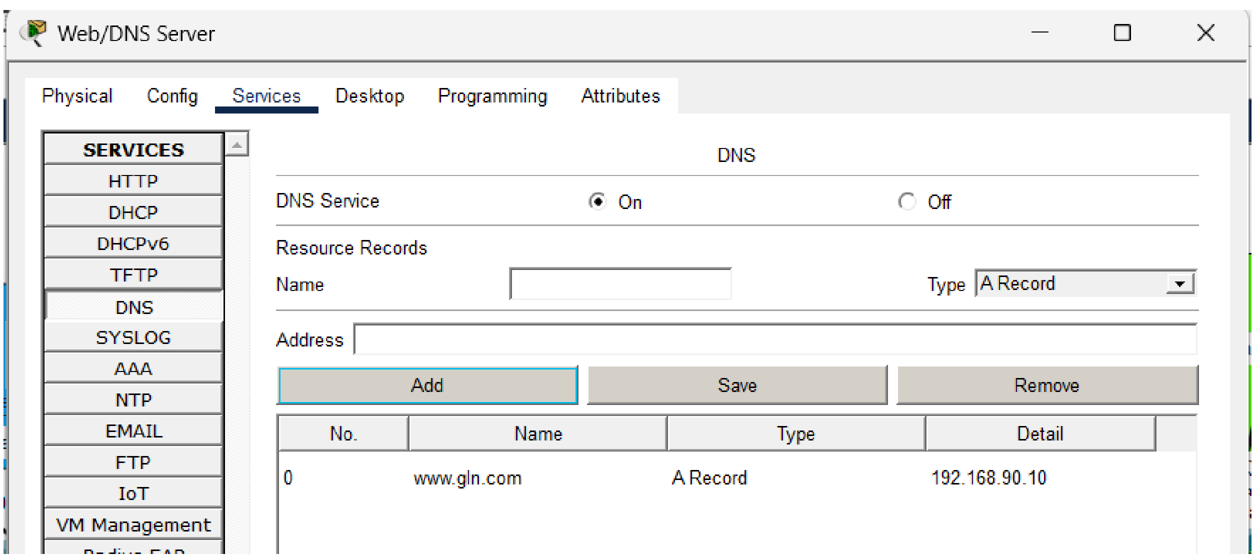
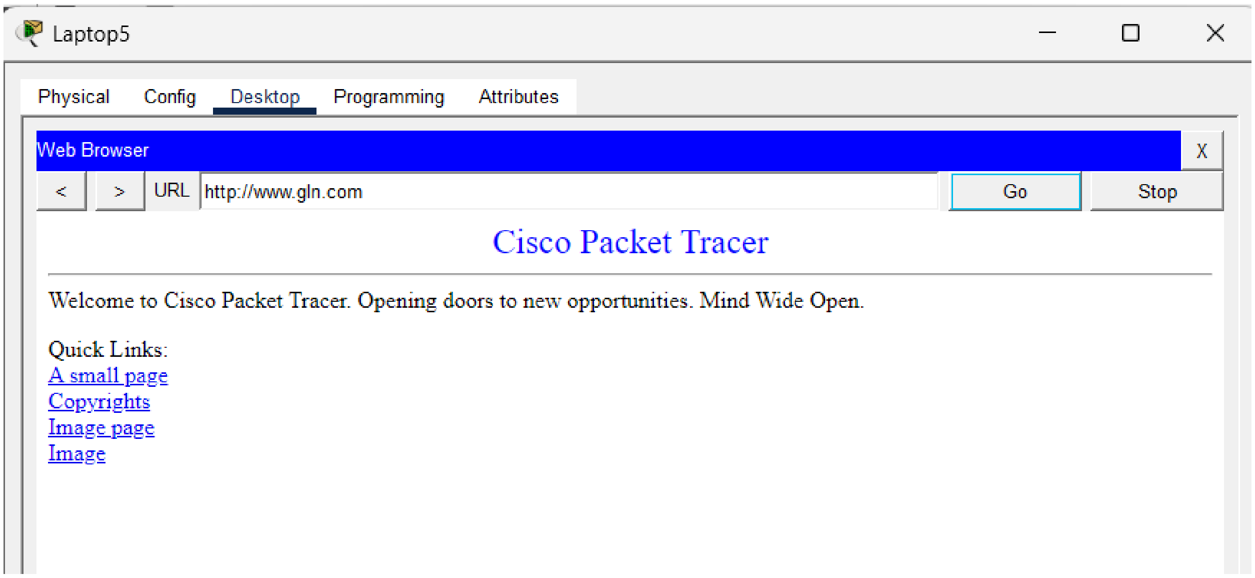
DNS Configuration:

- DNS server located in Data Center
- Static IP : 192.168.90.10
- DNS record creating





- Tested using web browser from remote sites



ACL Configuration:

- Block warehouse inventory (192.168.70.0) from accessing Data Center servers

```
R5-DC>en
Password:
R5-DC#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R5-DC(config)#ip access-list extended BLOCK-Inventory
R5-DC(config-ext-nacl)#deny ip 192.168.70.0 0.0.0.255 192.168.90.0 0.0.0.255
R5-DC(config-ext-nacl)#permit ip any any
R5-DC(config-ext-nacl)#interface gi0/2.90
R5-DC(config-subif)#ip access-group BLOCK-Inventory out
R5-DC(config-subif)#exit
R5-DC(config)#
```

Testing and Verification :

- Pinging from Admin laptop to Warehouse Shipping PC

```
C:\>ping 192.168.80.13

Pinging 192.168.80.13 with 32 bytes of data:

Reply from 192.168.80.13: bytes=32 time=17ms TTL=125
Reply from 192.168.80.13: bytes=32 time=3ms TTL=125
Reply from 192.168.80.13: bytes=32 time=2ms TTL=125
Reply from 192.168.80.13: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.80.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 17ms, Average = 6ms

C:\>
```

- Pinging from Inventory PC to the DNS server

```
C:\>ping 192.168.90.10

Pinging 192.168.90.10 with 32 bytes of data:

Reply from 172.16.0.13: Destination host unreachable.
Reply from 172.16.0.13: Destination host unreachable.
Reply from 172.16.0.13: Destination host unreachable.
Reply from 172.16.0.13: Destination host unreachable.

Ping statistics for 192.168.90.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

The Warehouse Inventory cannot ping the DNS server

- Pinging from another branch PC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.90.10

Pinging 192.168.90.10 with 32 bytes of data:

Reply from 192.168.90.10: bytes=32 time=2ms TTL=125
Reply from 192.168.90.10: bytes=32 time=21ms TTL=125
Reply from 192.168.90.10: bytes=32 time=22ms TTL=125
Reply from 192.168.90.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.90.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 22ms, Average = 11ms

C:\>
```

The DNS server is reachable from everyone except pf the warehouse inventory

- Checking the EIGRP routing

```
     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D       10.0.0.0/30 [90/2681856] via 10.0.0.17, 00:02:09, Serial0/0/0
D       10.0.0.4/30 [90/3193856] via 10.0.0.14, 00:02:09, Serial0/0/1
                    [90/3193856] via 10.0.0.17, 00:02:09, Serial0/0/0
D       10.0.0.8/30 [90/2681856] via 10.0.0.14, 00:02:09, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/1
L       10.0.0.13/32 is directly connected, Serial0/0/1
C       10.0.0.16/30 is directly connected, Serial0/0/0
L       10.0.0.18/32 is directly connected, Serial0/0/0
D    192.168.10.0/24 [90/2172416] via 10.0.0.17, 00:02:09, Serial0/0/0
D    192.168.20.0/24 [90/2172416] via 10.0.0.17, 00:02:09, Serial0/0/0
D    192.168.30.0/24 [90/2684416] via 10.0.0.17, 00:02:09, Serial0/0/0
D    192.168.40.0/24 [90/2684416] via 10.0.0.17, 00:02:09, Serial0/0/0
D    192.168.50.0/24 [90/2684416] via 10.0.0.14, 00:02:09, Serial0/0/1
D    192.168.60.0/24 [90/2684416] via 10.0.0.14, 00:02:09, Serial0/0/1
D    192.168.70.0/24 [90/2172416] via 10.0.0.14, 00:02:09, Serial0/0/1
D    192.168.80.0/24 [90/2172416] via 10.0.0.14, 00:02:09, Serial0/0/1
     192.168.90.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.90.0/24 is directly connected, GigabitEthernet0/2.90
L       192.168.90.1/32 is directly connected, GigabitEthernet0/2.90
     192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.99.0/24 is directly connected, GigabitEthernet0/2.99
L       192.168.99.1/32 is directly connected, GigabitEthernet0/2.99
```

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- Checking VLANs

Branch A

```
SW2-BrA#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
30   Sales                            active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
40   Support                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2-BrA#
```

Branch B

```
SW3-BrB#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
50   Marketing                        active    Fa0/2, Fa0/3, Fa0/4
60   Dev                              active    Fa0/5, Fa0/6, Fa0/7
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW3-BrB#
```

Warehouse

```
SW4-Whse#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
70   Inventory                        active    Fa0/2, Fa0/3, Fa0/4
80   Shipping                         active    Fa0/5, Fa0/6, Fa0/7
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW4-Whse#
```

Data Center

```
SW5-DC#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
90   Servers                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
99   management                       active    Fa0/6, Fa0/7, Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW5-DC#
```